

text Decision Log — Monday.com BI Agent (Skylark Drones)

1. Key Assumptions Made - Calendar Quarters & Timelines: Standard calendar quarterly periods are assumed (e.g., Q1 = Jan–Mar). Past-due close dates on active pipeline deals are flagged as data hygiene risks rather than automatically excluded. - Pipeline vs. Revenue Realization: "Deal Value" on the Deals board represents uncommitted potential pipeline revenue. "Contract Value" and "Billed Revenue" on the Work Orders board represent actual committed operational revenue. - Null & Missing Value Strategy: Missing owners, dates, or sectors are explicitly filled with "Unassigned" or "Unknown" labels during Pandas normalization. This preserves total row counts and ensures metrics aren't omitted during aggregation.

2. Trade-Offs Chosen & Why - Live GraphQL API Queries vs. Local SQL Database: Querying Monday.com directly via GraphQL on every manual sync ensures 100% real-time data parity. This avoids setting up and syncing an external persistent database (like PostgreSQL) within the sprint constraints. - Streamlit Community Cloud Hosting: Selected for zero-cost deployment, built-in session state management, and seamless GitHub integration. - In-Memory Data Transformation with Pandas: Normalizing numeric strings, stripping currency formatting (₹, \$), and filtering data in-memory keeps latency low and removes database maintenance overhead.

3. Data Resilience & Quality Strategy - Currency & Text Sanitization: Automated regex patterns strip non-numeric characters from monetary fields, preventing string-concatenation errors during mathematical aggregations. - Token Window Optimization: Large tables are capped/summarized before sending to the Gemini model to avoid token limit errors and maintain rapid response times. - Automated Caveat Reporting: Prompts instruct the model to explicitly scan for and append "Data Quality Warnings" (such as missing close dates, single-deal concentration risks, or stalled work orders) to every executive response.

4. "Leadership Updates" Interpretation - Executive Briefing Format: Formatted to deliver direct, bulleted answers first, followed by cross-board pipeline metrics. - Actionable Strategic Advice: Conclusions are provided under a dedicated "Leadership Brief" heading, highlighting bottleneck stages, high-value deal risks, and unstarted work orders for immediate founder intervention.